

EX:NO:4

PROGRAM CODE:

```
#include<stdio.h>

int main()
{
int bt[20],p[20],wt[20],tat[20],i,j,n,total=0,pos,temp;
float avg_wt,avg_tat;
printf("Enter number of process:");
scanf("%d",&n);
printf("\nEnter Burst Time:\n");
for(i=0;i<n;i++)
{
printf("p%d:",i+1);
scanf("%d",&bt[i]);
p[i]=i+1;
}
for(i=0;i<n;i++)
{
pos=i;
for(j=i+1;j<n;j++)
{
if(bt[j]<bt[pos]) pos=j;
}
temp=bt[i];
bt[i]=bt[pos];
bt[pos]=temp;
temp=p[i];
p[i]=p[pos];
p[pos]=temp;
}
wt[0]=0;
for(i=1;i<n;i++)
```

```

{
    wt[i]=0;
    for(j=0;j<i;j++)
        wt[i]+=bt[j];
    total+=wt[i];
}
avg_wt=(float)total/n;
total=0;
printf("\nProcess\tBurst Time\tWaiting Time\tTurnaround Time\n");
for(i=0;i<n;i++)
{
    tat[i]=bt[i]+wt[i]; total+=tat[i];
    printf("\n p%d\t %d\t %d\t %d",p[i],bt[i],wt[i],tat[i]);
}
avg_tat=(float)total/n;
printf("\n\nAverage Waiting Time=%f",avg_wt);
printf("\n\nAverage Turnaround Time=%f",avg_tat);
}

```

Output

```
Enter number of process: 4
Enter Burst Time:
P1: 4
P2: 6
P3: 7
P4: 5
P BT WT TAT
P1 4 0 4
P4 5 4 9
P2 6 9 15
P3 7 15 22
Average Waiting Time= 7.000000
AverageTurnaround Time= 12.500000

=== Code Execution Successful ===
```